

Benchmark results — Persistences, BLEND, AR-OLS, and ELM cost-function variants

The runs use $N_{\text{hidden}} = 500$ and $N_{\text{draws}} = 100$, $\approx 35\,000$ samples (2 years, August 2020 – August 2022), without dimensionality reduction.

The ELM and AR-OLS inputs are the standardized window of the $LB = 48$ lags, augmented by 4 cyclical time features (sin/cos encoding of the hour of the day and of the day of the year) for the target time step.

Predictions are clipped to $[0, +\infty)$ because the models sometimes predicted negative values, which are physically impossible.

Most ELM variants select their hyperparameter(s) from a small grid by validation RMSE (20 % of the training set). The λ column reports the selected value (or the selected tuple for two-parameter variants, in the order stated in each variant’s script). **Bold** marks the best NICE ^{Σ} among the cyclic ELM variants for each table (ties bolded together).

Two BLEND variants. **BLEND**_{optim} (*optimized*): λ per phase learned by directly minimizing the MSE on the training set (closed-form OLS solution). **BLEND**_{corr}: λ derived from the phase-conditioned empirical correlation ($\lambda = (1 + \rho)/2$). Both variants use the same persistences P and P° .

All data vs. daytime only. Each horizon is reported twice: *All data* includes all 30 min test steps (nights, where $P \approx 0$, dominate in count) and Daytime only, which restricts the evaluation to steps where the solar elevation at Palaiseau (48.71°N, 2.22°E) exceeds 0°.

FH = 0.5 h

Table 1: All data – $FH = 0.5$ h

Method	N_θ	N_h	N_d	λ	RMSE (W)	nRMSE	nMBE	nMAE	R^2	NICE ¹	NICE ²	NICE ³	NICE ^{Σ}
Persistence P	0	–	–	–	1169.0	0.490	~ 0	0.239	0.910	1.000	1.000	1.000	1.000
Persistence P [°]	0	–	–	–	2298.3	0.963	~ 0	0.429	0.653	1.793	1.966	1.938	1.899
BLEND _{corr}	0	–	–	–	1133.1	0.475	−0.002	0.233	0.916	0.974	0.969	0.967	0.970
BLEND _{optim}	48	–	–	–	1079.7	0.452	−0.014	0.216	0.923	0.901	0.924	0.931	0.918
AR-OLS	53	–	–	–	1012.4	0.424	+0.014	0.212	0.933	0.887	0.866	0.880	0.878
ELM (OLS)	27000	500	100	–	991.6	0.415	+0.021	0.212	0.935	0.884	0.848	0.861	0.865
ELM (Ridge)	27000	500	100	25	986.8	0.413	+0.022	0.210	0.936	0.879	0.844	0.859	0.861
ELM (Rob. Risk)	27000	500	100	5	986.8	0.413	+0.022	0.210	0.936	0.879	0.844	0.859	0.861
ELM (Tikhonov)	27000	500	100	25	987.9	0.414	+0.024	0.211	0.936	0.881	0.845	0.860	0.862
ELM (Box-Cox)	27000	500	100	(25,1)	986.8	0.413	+0.022	0.210	0.936	0.879	0.844	0.859	0.861
ELM (MAE)	27000	500	100	(10,10)	983.7	0.412	+0.020	0.205	0.936	0.859	0.841	0.862	0.854
ELM (Huber)	27000	500	100	25	983.3	0.412	+0.019	0.201	0.936	0.841	0.841	0.868	0.850
ELM (L3)	27000	500	100	(10,500)	1161.9	0.487	+0.050	0.288	0.911	1.202	0.994	0.936	1.044
ELM (Corr)	27000	500	100	(25,0.5)	997.6	0.418	+0.022	0.201	0.935	0.841	0.853	0.892	0.862
ELM (Elastic Net)	27000	500	100	(1,25)	986.8	0.413	+0.022	0.210	0.936	0.879	0.844	0.859	0.861
ELM (M-Est.)	27000	500	100	25	985.0	0.413	+0.019	0.200	0.936	0.835	0.843	0.871	0.850
ELM (Lp)	27000	500	100	(25,1.5)	982.9	0.412	+0.021	0.205	0.937	0.855	0.841	0.863	0.853
ELM (GLM)	27000	500	100	10	1115.3	0.467	+0.085	0.274	0.918	1.144	0.954	0.940	1.013
ELM _{no cyc} (OLS)	25000	500	100	–	1000.2	0.419	+0.020	0.211	0.934	0.881	0.856	0.873	0.870
ELM _{no cyc} (Ridge)	25000	500	100	25	992.6	0.416	+0.016	0.208	0.935	0.869	0.849	0.866	0.861

Table 2: Daytime only ($\theta_{\text{sol}} > 0^\circ$) – $FH = 0.5$ h

Method	N_θ	N_h	N_d	λ	RMSE (W)	nRMSE	nMBE	nMAE	R^2	NICE ¹	NICE ²	NICE ³	NICE ^{Σ}
Persistence P	0	–	–	–	1650.0	0.347	~ 0	0.239	0.857	1.000	1.000	1.000	1.000
Persistence P ^o	0	–	–	–	3244.1	0.682	~ 0	0.429	0.448	1.795	1.966	1.938	1.900
BLEND _{corr}	0	–	–	–	1599.3	0.336	−0.002	0.233	0.866	0.974	0.969	0.967	0.970
BLEND _{optim}	48	–	–	–	1524.0	0.320	−0.014	0.216	0.878	0.902	0.924	0.931	0.919
AR-OLS	53	–	–	–	1423.3	0.299	+0.001	0.200	0.894	0.836	0.863	0.880	0.859
ELM (OLS)	27000	500	100	–	1382.8	0.291	−0.001	0.189	0.900	0.793	0.838	0.860	0.830
ELM (Ridge)	27000	500	100	25	1379.4	0.290	+0.001	0.190	0.900	0.793	0.836	0.858	0.829
ELM (Rob. Risk)	27000	500	100	5	1379.4	0.290	+0.001	0.190	0.900	0.793	0.836	0.858	0.829
ELM (Tikhonov)	27000	500	100	25	1379.4	0.290	+0.002	0.189	0.900	0.791	0.836	0.859	0.829
ELM (Box-Cox)	27000	500	100	(25,1)	1379.4	0.290	+0.001	0.190	0.900	0.793	0.836	0.858	0.829
ELM (MAE)	27000	500	100	(10,10)	1376.6	0.289	+0.001	0.186	0.901	0.780	0.834	0.861	0.825
ELM (Huber)	27000	500	100	25	1377.6	0.290	+0.002	0.184	0.900	0.768	0.835	0.868	0.823
ELM (L3)	27000	500	100	(10,500)	1576.5	0.331	+0.001	0.238	0.870	0.995	0.955	0.927	0.959
ELM (Corr)	27000	500	100	(25,0.5)	1394.6	0.293	+0.002	0.181	0.898	0.759	0.845	0.891	0.832
ELM (Elastic Net)	27000	500	100	(1,25)	1379.4	0.290	+0.001	0.190	0.900	0.793	0.836	0.858	0.829
ELM (M-Est.)	27000	500	100	25	1381.0	0.290	+0.002	0.183	0.900	0.766	0.837	0.871	0.825
ELM (Lp)	27000	500	100	(25,1.5)	1375.6	0.289	+0.002	0.186	0.901	0.777	0.834	0.862	0.824
ELM (GLM)	27000	500	100	10	1532.9	0.322	+0.035	0.224	0.877	0.936	0.929	0.936	0.934
ELM _{no cyc} (OLS)	25000	500	100	–	1397.1	0.294	−0.001	0.190	0.898	0.796	0.847	0.872	0.838
ELM _{no cyc} (Ridge)	25000	500	100	25	1390.8	0.292	−0.002	0.190	0.899	0.797	0.843	0.866	0.835

FH = 1 hTable 3: All data – $FH = 1$ h

Method	N_θ	N_h	N_d	λ	RMSE (W)	nRMSE	nMBE	nMAE	R^2	NICE ¹	NICE ²	NICE ³	NICE ^{Σ}
Persistence P	0	–	–	–	1699.5	0.712	~ 0	0.379	0.810	1.000	1.000	1.000	1.000
Persistence P ^o	0	–	–	–	2298.3	0.963	~ 0	0.429	0.653	1.133	1.352	1.429	1.305
BLEND _{corr}	0	–	–	–	1603.5	0.672	−0.004	0.357	0.831	0.941	0.944	0.945	0.943
BLEND _{optim}	48	–	–	–	1379.6	0.578	−0.031	0.283	0.875	0.746	0.812	0.848	0.802
AR-OLS	53	–	–	–	1277.0	0.535	+0.022	0.285	0.893	0.753	0.751	0.785	0.763
ELM (OLS)	27000	500	100	–	1207.0	0.506	+0.024	0.269	0.904	0.709	0.710	0.751	0.724
ELM (Ridge)	27000	500	100	25	1202.0	0.503	+0.019	0.266	0.905	0.701	0.707	0.748	0.719
ELM (Rob. Risk)	27000	500	100	5	1202.0	0.503	+0.019	0.266	0.905	0.701	0.707	0.748	0.719
ELM (Tikhonov)	27000	500	100	25	1203.0	0.504	+0.021	0.267	0.905	0.703	0.708	0.749	0.720
ELM (Box-Cox)	27000	500	100	(10,0.75)	1197.6	0.502	−0.019	0.246	0.906	0.648	0.705	0.756	0.703
ELM (MAE)	27000	500	100	(25,10)	1200.9	0.503	+0.015	0.259	0.905	0.683	0.707	0.753	0.714
ELM (Huber)	27000	500	100	25	1203.1	0.504	+0.016	0.255	0.905	0.672	0.708	0.760	0.713
ELM (L3)	27000	500	100	(10,500)	1380.7	0.578	+0.053	0.356	0.875	0.939	0.812	0.796	0.849
ELM (Corr)	27000	500	100	(25,0.5)	1213.1	0.508	+0.027	0.262	0.903	0.692	0.714	0.771	0.726
ELM (Elastic Net)	27000	500	100	(10,25)	1202.0	0.503	+0.019	0.266	0.905	0.701	0.707	0.748	0.719
ELM (M-Est.)	27000	500	100	10	1214.2	0.509	+0.015	0.253	0.903	0.669	0.714	0.770	0.718
ELM (Lp)	27000	500	100	(25,1.25)	1199.7	0.502	+0.016	0.258	0.905	0.682	0.706	0.753	0.714
ELM (GLM)	27000	500	100	10	1288.5	0.540	+0.062	0.303	0.891	0.798	0.758	0.797	0.784
ELM _{no cyc} (OLS)	25000	500	100	–	1223.8	0.513	+0.021	0.270	0.902	0.712	0.720	0.763	0.732
ELM _{no cyc} (Ridge)	25000	500	100	25	1218.1	0.510	+0.016	0.266	0.903	0.702	0.717	0.762	0.727

Table 4: Daytime only ($\theta_{\text{sol}} > 0^\circ$) – $FH = 1$ h

Method	N_θ	N_h	N_d	λ	RMSE (W)	nRMSE	nMBE	nMAE	R^2	NICE ¹	NICE ²	NICE ³	NICE ^{Σ}
Persistence P	0	–	–	–	2397.6	0.504	−0.003	0.376	0.698	1.000	1.000	1.000	1.000
Persistence P ^o	0	–	–	–	3244.1	0.682	~ 0	0.429	0.448	1.140	1.353	1.429	1.308
BLEND _{corr}	0	–	–	–	2262.4	0.476	−0.006	0.354	0.731	0.941	0.944	0.945	0.943
BLEND _{optim}	48	–	–	–	1947.2	0.409	−0.031	0.282	0.801	0.750	0.812	0.848	0.804
AR-OLS	53	–	–	–	1792.6	0.377	+0.004	0.267	0.831	0.709	0.748	0.784	0.747
ELM (OLS)	27000	500	100	–	1682.6	0.354	−0.004	0.241	0.851	0.639	0.702	0.750	0.697
ELM (Ridge)	27000	500	100	25	1681.2	0.353	−0.005	0.241	0.852	0.641	0.701	0.747	0.697
ELM (Rob. Risk)	27000	500	100	5	1681.2	0.353	−0.005	0.241	0.852	0.641	0.701	0.747	0.697
ELM (Tikhonov)	27000	500	100	25	1680.3	0.353	−0.005	0.241	0.852	0.639	0.701	0.748	0.696
ELM (Box-Cox)	27000	500	100	(10,0.75)	1687.5	0.355	−0.029	0.236	0.851	0.627	0.704	0.756	0.696
ELM (MAE)	27000	500	100	(25,10)	1683.2	0.354	−0.005	0.238	0.851	0.632	0.702	0.753	0.696
ELM (Huber)	27000	500	100	25	1686.6	0.355	−0.005	0.234	0.851	0.622	0.703	0.760	0.695
ELM (L3)	27000	500	100	(10,500)	1877.3	0.395	−0.005	0.298	0.815	0.791	0.783	0.789	0.787
ELM (Corr)	27000	500	100	(25,0.5)	1692.9	0.356	~ 0	0.235	0.850	0.624	0.706	0.770	0.700
ELM (Elastic Net)	27000	500	100	(10,25)	1681.2	0.353	−0.005	0.241	0.852	0.641	0.701	0.747	0.697
ELM (M-Est.)	27000	500	100	10	1704.6	0.358	−0.003	0.235	0.848	0.625	0.711	0.770	0.702
ELM (Lp)	27000	500	100	(25,1.25)	1681.1	0.353	−0.005	0.237	0.852	0.630	0.701	0.753	0.695
ELM (GLM)	27000	500	100	10	1796.5	0.378	+0.023	0.264	0.831	0.702	0.749	0.796	0.749
ELM _{no cyc} (OLS)	25000	500	100	–	1709.8	0.360	−0.005	0.244	0.847	0.649	0.713	0.762	0.708
ELM _{no cyc} (Ridge)	25000	500	100	25	1707.6	0.359	−0.006	0.244	0.847	0.649	0.712	0.762	0.708

FH = 3 hTable 5: All data – $FH = 3$ h

Method	N_θ	N_h	N_d	λ	RMSE (W)	nRMSE	nMBE	nMAE	R^2	NICE ¹	NICE ²	NICE ³	NICE ^{Σ}
Persistence P	0	–	–	–	3594.0	1.505	−0.001	0.890	0.151	1.000	1.000	1.000	1.000
Persistence P ^o	0	–	–	–	2298.5	0.963	~ 0	0.429	0.653	0.483	0.640	0.727	0.616
BLEND _{corr}	0	–	–	–	3007.6	1.260	−0.002	0.728	0.406	0.818	0.837	0.846	0.834
BLEND _{optim}	48	–	–	–	1914.3	0.802	−0.072	0.383	0.759	0.430	0.533	0.597	0.520
AR-OLS	53	–	–	–	1800.2	0.754	+0.029	0.425	0.787	0.478	0.501	0.540	0.506
ELM (OLS)	27000	500	100	–	1665.3	0.697	+0.026	0.385	0.818	0.433	0.463	0.506	0.468
ELM (Ridge)	27000	500	100	25	1660.3	0.695	+0.017	0.382	0.819	0.429	0.462	0.505	0.465
ELM (Rob. Risk)	27000	500	100	5	1660.3	0.695	+0.017	0.382	0.819	0.429	0.462	0.505	0.465
ELM (Tikhonov)	27000	500	100	25	1660.6	0.695	+0.021	0.383	0.819	0.430	0.462	0.505	0.466
ELM (Box-Cox)	27000	500	100	(25,0.75)	1663.4	0.697	−0.043	0.357	0.818	0.401	0.463	0.513	0.459
ELM (MAE)	27000	500	100	(25,100)	1673.6	0.701	+0.005	0.368	0.816	0.413	0.466	0.517	0.465
ELM (Huber)	27000	500	100	25	1679.3	0.703	+0.005	0.364	0.815	0.409	0.467	0.521	0.466
ELM (L3)	27000	500	100	(10,500)	1822.3	0.763	+0.081	0.483	0.782	0.542	0.507	0.518	0.522
ELM (Corr)	27000	500	100	(25,0.5)	1660.9	0.696	+0.032	0.384	0.819	0.432	0.462	0.506	0.467
ELM (Elastic Net)	27000	500	100	(1,25)	1660.3	0.695	+0.017	0.382	0.819	0.429	0.462	0.505	0.465
ELM (M-Est.)	27000	500	100	25	1699.6	0.712	−0.003	0.360	0.810	0.404	0.473	0.531	0.470
ELM (Lp)	27000	500	100	(25,1.25)	1670.6	0.700	+0.010	0.370	0.817	0.416	0.465	0.515	0.465
ELM (GLM)	27000	500	100	10	1737.0	0.727	+0.056	0.405	0.802	0.455	0.483	0.535	0.491
ELM _{no cyc} (OLS)	25000	500	100	–	1713.1	0.717	+0.020	0.392	0.807	0.441	0.477	0.523	0.480
ELM _{no cyc} (Ridge)	25000	500	100	25	1705.2	0.714	+0.012	0.388	0.809	0.436	0.474	0.521	0.477

Table 6: Daytime only ($\theta_{\text{sol}} > 0^\circ$) – $FH = 3 \text{ h}$

Method	N_θ	N_h	N_d	λ	RMSE (W)	nRMSE	nMBE	nMAE	R^2	NICE ¹	NICE ²	NICE ³	NICE ^{Σ}
Persistence P	0	–	–	–	4919.5	1.034	−0.082	0.808	−0.270	1.000	1.000	1.000	1.000
Persistence P ^o	0	–	–	–	3244.4	0.682	~ 0	0.429	0.448	0.531	0.660	0.738	0.643
BLEND _{corr}	0	–	–	–	4147.1	0.872	−0.059	0.671	0.098	0.831	0.843	0.850	0.841
BLEND _{optim}	48	–	–	–	2701.8	0.568	−0.073	0.382	0.617	0.473	0.549	0.606	0.543
AR-OLS	53	–	–	–	2520.8	0.530	−0.001	0.395	0.667	0.489	0.512	0.548	0.516
ELM (OLS)	27000	500	100	–	2326.0	0.489	−0.010	0.349	0.716	0.433	0.473	0.513	0.473
ELM (Ridge)	27000	500	100	25	2325.3	0.489	−0.014	0.351	0.716	0.434	0.473	0.512	0.473
ELM (Rob. Risk)	27000	500	100	5	2325.3	0.489	−0.014	0.351	0.716	0.434	0.473	0.512	0.473
ELM (Tikhonov)	27000	500	100	25	2323.3	0.488	−0.012	0.350	0.717	0.433	0.472	0.512	0.472
ELM (Box-Cox)	27000	500	100	(25,0.75)	2343.8	0.493	−0.055	0.344	0.712	0.426	0.476	0.520	0.474
ELM (MAE)	27000	500	100	(25,100)	2350.0	0.494	−0.020	0.342	0.710	0.424	0.478	0.524	0.475
ELM (Huber)	27000	500	100	25	2357.8	0.496	−0.020	0.338	0.708	0.419	0.479	0.529	0.476
ELM (L3)	27000	500	100	(10,500)	2487.8	0.523	+0.008	0.410	0.675	0.507	0.506	0.521	0.511
ELM (Corr)	27000	500	100	(25,0.5)	2316.0	0.487	−0.006	0.346	0.719	0.428	0.471	0.512	0.471
ELM (Elastic Net)	27000	500	100	(1,25)	2325.3	0.489	−0.014	0.351	0.716	0.434	0.473	0.512	0.473
ELM (M-Est.)	27000	500	100	25	2388.8	0.502	−0.026	0.337	0.701	0.417	0.486	0.539	0.481
ELM (Lp)	27000	500	100	(25,1.25)	2344.2	0.493	−0.017	0.343	0.712	0.425	0.477	0.522	0.475
ELM (GLM)	27000	500	100	10	2433.9	0.512	+0.014	0.363	0.689	0.449	0.495	0.543	0.496
ELM _{no cyc} (OLS)	25000	500	100	–	2396.0	0.504	−0.016	0.356	0.699	0.441	0.487	0.530	0.486
ELM _{no cyc} (Ridge)	25000	500	100	25	2390.1	0.502	−0.019	0.356	0.700	0.441	0.486	0.529	0.485

FH = 6 hTable 7: All data – $FH = 6 \text{ h}$

Method	N_θ	N_h	N_d	λ	RMSE (W)	nRMSE	nMBE	nMAE	R^2	NICE ¹	NICE ²	NICE ³	NICE ^{Σ}
Persistence P	0	–	–	–	5509.8	2.307	~ 0	1.510	−0.994	1.000	1.000	1.000	1.000
Persistence P ^o	0	–	–	–	2298.7	0.963	~ 0	0.429	0.653	0.284	0.417	0.494	0.398
BLEND _{corr}	0	–	–	–	3817.0	1.598	−0.005	1.017	0.043	0.673	0.693	0.706	0.691
BLEND _{optim}	48	–	–	–	2164.7	0.906	−0.115	0.423	0.692	0.280	0.393	0.463	0.379
AR-OLS	53	–	–	–	1960.5	0.821	+0.017	0.458	0.748	0.303	0.356	0.402	0.354
ELM (OLS)	27000	500	100	–	1879.7	0.787	+0.014	0.436	0.768	0.289	0.341	0.385	0.338
ELM (Ridge)	27000	500	100	25	1859.9	0.779	+0.007	0.425	0.773	0.282	0.338	0.382	0.334
ELM (Rob. Risk)	27000	500	100	10	1863.4	0.780	+0.001	0.425	0.772	0.281	0.338	0.383	0.334
ELM (Tikhonov)	27000	500	100	25	1871.9	0.784	+0.010	0.432	0.770	0.286	0.340	0.384	0.336
ELM (Box-Cox)	27000	500	100	(25,0.75)	1870.5	0.783	−0.055	0.405	0.770	0.268	0.339	0.387	0.332
ELM (MAE)	27000	500	100	(25,100)	1883.5	0.789	−0.005	0.413	0.767	0.274	0.342	0.393	0.336
ELM (Huber)	27000	500	100	25	1895.0	0.793	−0.005	0.410	0.764	0.272	0.344	0.397	0.338
ELM (L3)	27000	500	100	(10,500)	2027.2	0.849	+0.064	0.530	0.730	0.351	0.368	0.389	0.369
ELM (Corr)	27000	500	100	(25,0.5)	1886.4	0.790	+0.020	0.439	0.766	0.291	0.342	0.386	0.340
ELM (Elastic Net)	27000	500	100	(10,25)	1859.9	0.779	+0.007	0.425	0.773	0.282	0.338	0.382	0.334
ELM (M-Est.)	27000	500	100	25	1916.2	0.802	−0.011	0.406	0.759	0.269	0.348	0.405	0.340
ELM (Lp)	27000	500	100	(25,1.25)	1878.4	0.787	~ 0	0.416	0.768	0.275	0.341	0.391	0.336
ELM (GLM)	27000	500	100	10	1935.2	0.810	+0.060	0.458	0.754	0.303	0.351	0.398	0.351
ELM _{no cyc} (OLS)	25000	500	100	–	1930.9	0.809	+0.007	0.448	0.755	0.297	0.350	0.398	0.348
ELM _{no cyc} (Ridge)	25000	500	100	25	1909.3	0.799	−0.002	0.439	0.761	0.291	0.347	0.393	0.343

Table 8: Daytime only ($\theta_{\text{sol}} > 0^\circ$) – $FH = 6$ h

Method	N_θ	N_h	N_d	λ	RMSE (W)	nRMSE	nMBE	nMAE	R^2	NICE ¹	NICE ²	NICE ³	NICE ^{Σ}
Persistence P	0	–	–	–	6756.5	1.420	−0.408	1.103	−1.395	1.000	1.000	1.000	1.000
Persistence P ^o	0	–	–	–	3244.6	0.682	~ 0	0.429	0.448	0.389	0.480	0.539	0.469
BLEND _{corr}	0	–	–	–	4911.6	1.032	−0.235	0.788	−0.266	0.714	0.727	0.735	0.725
BLEND _{optim}	48	–	–	–	3055.4	0.642	−0.115	0.423	0.510	0.383	0.452	0.505	0.447
AR-OLS	53	–	–	–	2750.6	0.578	−0.014	0.427	0.603	0.387	0.407	0.438	0.411
ELM (OLS)	27000	500	100	–	2631.2	0.553	−0.022	0.400	0.637	0.362	0.389	0.419	0.390
ELM (Ridge)	27000	500	100	25	2610.1	0.549	−0.023	0.396	0.643	0.359	0.386	0.417	0.387
ELM (Rob. Risk)	27000	500	100	10	2618.0	0.550	−0.025	0.399	0.640	0.362	0.387	0.418	0.389
ELM (Tikhonov)	27000	500	100	25	2624.1	0.552	−0.023	0.398	0.639	0.361	0.388	0.418	0.389
ELM (Box-Cox)	27000	500	100	(25,0.75)	2636.5	0.554	−0.068	0.392	0.635	0.356	0.390	0.422	0.389
ELM (MAE)	27000	500	100	(25,100)	2648.4	0.557	−0.029	0.389	0.632	0.352	0.392	0.428	0.391
ELM (Huber)	27000	500	100	25	2664.0	0.560	−0.030	0.385	0.628	0.349	0.394	0.433	0.392
ELM (L3)	27000	500	100	(10,500)	2782.0	0.585	−0.006	0.460	0.594	0.417	0.412	0.420	0.416
ELM (Corr)	27000	500	100	(25,0.5)	2635.9	0.554	−0.020	0.400	0.635	0.362	0.390	0.420	0.391
ELM (Elastic Net)	27000	500	100	(10,25)	2610.1	0.549	−0.023	0.396	0.643	0.359	0.386	0.417	0.387
ELM (M-Est.)	27000	500	100	25	2696.0	0.567	−0.034	0.383	0.619	0.347	0.399	0.441	0.396
ELM (Lp)	27000	500	100	(25,1.25)	2639.8	0.555	−0.026	0.390	0.634	0.353	0.391	0.426	0.390
ELM (GLM)	27000	500	100	10	2712.4	0.570	+0.014	0.413	0.614	0.374	0.401	0.433	0.403
ELM _{no cyc} (OLS)	25000	500	100	–	2704.2	0.568	−0.031	0.409	0.616	0.371	0.400	0.433	0.402
ELM _{no cyc} (Ridge)	25000	500	100	25	2679.8	0.563	−0.035	0.406	0.623	0.368	0.397	0.429	0.398

FH = 10 hTable 9: All data – $FH = 10$ h

Method	N_θ	N_h	N_d	λ	RMSE (W)	nRMSE	nMBE	nMAE	R^2	NICE ¹	NICE ²	NICE ³	NICE ^{Σ}
Persistence P	0	–	–	–	6403.1	2.680	~ 0	1.927	−1.693	1.000	1.000	1.000	1.000
Persistence P ^o	0	–	–	–	2298.9	0.962	~ 0	0.429	0.653	0.223	0.359	0.440	0.341
BLEND _{corr}	0	–	–	–	3731.3	1.562	−0.004	1.110	0.086	0.576	0.583	0.592	0.584
BLEND _{optim}	48	–	–	–	2224.0	0.931	−0.134	0.433	0.675	0.225	0.347	0.423	0.332
AR-OLS	53	–	–	–	1964.9	0.823	+0.014	0.458	0.746	0.238	0.307	0.359	0.301
ELM (OLS)	27000	500	100	–	1894.3	0.793	+0.016	0.439	0.764	0.228	0.296	0.345	0.289
ELM (Ridge)	27000	500	100	25	1887.3	0.790	+0.008	0.434	0.766	0.225	0.295	0.344	0.288
ELM (Rob. Risk)	27000	500	100	10	1888.8	0.791	−0.003	0.432	0.766	0.224	0.295	0.344	0.288
ELM (Tikhonov)	27000	500	100	25	1889.3	0.791	+0.011	0.436	0.766	0.226	0.295	0.344	0.288
ELM (Box-Cox)	27000	500	100	(25,0.75)	1899.9	0.795	−0.061	0.412	0.763	0.214	0.297	0.349	0.286
ELM (MAE)	27000	500	100	(25,100)	1903.5	0.797	−0.011	0.417	0.762	0.217	0.297	0.352	0.289
ELM (Huber)	27000	500	100	25	1914.6	0.801	−0.011	0.414	0.759	0.215	0.299	0.356	0.290
ELM (L3)	27000	500	100	(10,500)	2078.3	0.870	+0.062	0.549	0.716	0.285	0.325	0.353	0.321
ELM (Corr)	27000	500	100	(25,0.5)	1900.7	0.796	+0.014	0.440	0.763	0.228	0.297	0.345	0.290
ELM (Elastic Net)	27000	500	100	(10,25)	1887.3	0.790	+0.008	0.434	0.766	0.225	0.295	0.344	0.288
ELM (M-Est.)	27000	500	100	25	1931.0	0.808	−0.017	0.410	0.755	0.213	0.302	0.361	0.292
ELM (Lp)	27000	500	100	(25,1.25)	1900.7	0.796	−0.006	0.420	0.763	0.218	0.297	0.351	0.289
ELM (GLM)	27000	500	100	10	1972.5	0.826	+0.051	0.458	0.744	0.238	0.308	0.362	0.302
ELM _{no cyc} (OLS)	25000	500	100	–	1948.3	0.816	+0.004	0.451	0.751	0.234	0.304	0.355	0.298
ELM _{no cyc} (Ridge)	25000	500	100	25	1940.2	0.812	−0.003	0.444	0.753	0.231	0.303	0.354	0.296

Table 10: Daytime only ($\theta_{\text{sol}} > 0^\circ$) – $FH = 10$ h

Method	N_θ	N_h	N_d	λ	RMSE (W)	nRMSE	nMBE	nMAE	R^2	NICE ¹	NICE ²	NICE ³	NICE ^{Σ}
Persistence P	0	–	–	–	6688.9	1.406	−0.852	1.074	−1.348	1.000	1.000	1.000	1.000
Persistence P ^o	0	–	–	–	3244.6	0.682	~ 0	0.429	0.448	0.399	0.485	0.540	0.475
BLEND _{corr}	0	–	–	–	4262.7	0.896	−0.441	0.672	0.047	0.626	0.637	0.649	0.637
BLEND _{optim}	48	–	–	–	3138.9	0.660	−0.135	0.433	0.483	0.403	0.469	0.519	0.464
AR-OLS	53	–	–	–	2758.3	0.580	−0.016	0.429	0.601	0.399	0.412	0.440	0.417
ELM (OLS)	27000	500	100	–	2653.2	0.558	−0.019	0.404	0.631	0.376	0.397	0.423	0.398
ELM (Ridge)	27000	500	100	25	2648.3	0.557	−0.022	0.404	0.632	0.376	0.396	0.422	0.398
ELM (Rob. Risk)	27000	500	100	10	2655.3	0.558	−0.027	0.407	0.630	0.379	0.397	0.422	0.399
ELM (Tikhonov)	27000	500	100	25	2649.2	0.557	−0.021	0.404	0.632	0.376	0.396	0.422	0.398
ELM (Box-Cox)	27000	500	100	(25,0.75)	2678.6	0.563	−0.072	0.401	0.624	0.373	0.400	0.428	0.401
ELM (MAE)	27000	500	100	(25,100)	2678.7	0.563	−0.033	0.396	0.624	0.368	0.400	0.432	0.400
ELM (Huber)	27000	500	100	25	2694.0	0.566	−0.033	0.392	0.619	0.365	0.403	0.437	0.402
ELM (L3)	27000	500	100	(10,500)	2845.0	0.598	−0.013	0.474	0.575	0.442	0.425	0.429	0.432
ELM (Corr)	27000	500	100	(25,0.5)	2662.7	0.560	−0.020	0.406	0.628	0.378	0.398	0.424	0.400
ELM (Elastic Net)	27000	500	100	(10,25)	2648.3	0.557	−0.022	0.404	0.632	0.376	0.396	0.422	0.398
ELM (M-Est.)	27000	500	100	25	2718.6	0.571	−0.037	0.390	0.612	0.363	0.406	0.444	0.404
ELM (Lp)	27000	500	100	(25,1.25)	2673.5	0.562	−0.029	0.397	0.625	0.369	0.400	0.431	0.400
ELM (GLM)	27000	500	100	10	2769.9	0.582	+0.015	0.422	0.597	0.393	0.414	0.444	0.417
ELM _{no cyc} (OLS)	25000	500	100	–	2734.2	0.575	−0.029	0.417	0.608	0.388	0.409	0.436	0.411
ELM _{no cyc} (Ridge)	25000	500	100	25	2726.5	0.573	−0.032	0.416	0.610	0.387	0.408	0.435	0.410